

Project Backlog, To-Dos, and Decision Log

North Star Architecture (v2 — GCP-native, no Databricks)

Source drops (CSV/JSON) → **GCS drop Dataflow** (ingest + normalize → JSONL) → **GCS raw Dataproc Spark** → Bronze → Silver (contract + cleaning + DQ + deadletter) → Gold (curated) Publish Gold → **BigQuery** Ops tables in **BigQuery**: `ops.pipeline_runs`, `ops.dq_results`, `ops.schema_registry` Part 2 (mandatory): **Vertex AI ADK** + **RAG Agents** (generate/validate cleaning & transforms) + **HITL gating** + versioning

Current Status

- ☒ Training Teams Flow doc updated and locked (row = prompt×step, reviewer edits per-curator, auto-rater step-level).
 - ☒ Sample source batches generated (auto JSON + human CSV).
 - ☒ Ingestion layer converts sources → **Raw JSONL** (meta + payload) and lands in **GCS raw**.
 - ☐ Databricks blocked by serverless/GCS access + workspace/compute limits → moving to Dataflow/Dataproc.
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MVP Backlog (Part 1)

Step 5 — Bronze (Dataproc)

- Read GCS raw JSONL → write Bronze (partitioned, minimal transforms).

Step 6 — Silver Contract (Schema)

- Define canonical silver schema + drift policy (extras column vs versioned tables).
- Persist contract + schema hashes into `ops.schema_registry`.

Step 7 — Silver Transforms (Cleaning)

- Normalize ratings/violations, enforce reviewer overrides, standardize step fields.
- Create unified view for human + auto-rater rows.

Step 8 — DQ + Deadletter

- Null/dup/row-count drift checks.
- Quarantine bad rows (GCS deadletter path + Delta/Parquet table).
- Write DQ outcomes to `ops.dq_results`.

Step 9 — Gold

- Gold Training datasets (SFT / multi-objective scores; preference if available).
- Gold Analytics datasets (disagreement, reviewer edit rate, violations, step-count distribution).

Step 10 — Publish to BigQuery

- Load Gold tables into BQ.
 - Validation checks + basic performance sanity.
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Part 2 (Mandatory) — Agents + Orchestration

- Orchestration: Cloud Scheduler + Workflows/Composer to trigger ingestion + dataproc jobs.
 - Vertex AI ADK agent:
 - RAG grounded in `schema_registry` + contracts + examples
 - generates cleaning/transform code
 - gated by tests/DQ + HITL approval
 - version prompts/corpus/output code
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Add-on Learning Mini Projects (Later)

Mini Project A (AWS) — Databricks + Snowflake Hands-On

Added per user request: After this GCP-native project, do a small AWS project to get hands-on with: - Databricks (Delta, jobs) - Snowflake (staging, warehouses, ELT) - S3 → Databricks → Snowflake → Gold marts - small scope, 1 dataset, 1-2 gold tables, CI/CD basics

Next To-Do List (Immediate)

1) **Step 5 Dataproc Bronze:** implement `bronze_ingest.py` + run via `gcloud dataproc jobs submit pyspark`. 2) Create `ops.schema_registry` table in BigQuery. 3) Draft Silver contract YAML/JSON for the row payload.

Decision Log

- **2026-02-09** — Removed Databricks from MVP execution due to serverless/GCS access friction + workspace limits. Adopted **Dataflow + Dataproc + BigQuery** for the core pipeline.
- **2026-02-09** — Planned a separate **AWS mini project** later for Databricks + Snowflake hands-on (kept out of main GCP project to avoid mixed-cloud confusion).